

Midterm Exam 3 Retake V3 Solutions & Rubric

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Suggested Point Distribution:

Undergraduate Total: 80 pts — Graduate Total: 100 pts

- **Exercise 1:** 20 points (5 pts for PFD, 5 pts per series region)
- **Exercise 2:** 20 points (5 pts for identifying poles, 10 pts for residues, 5 pts for final integral)
- **Exercise 3:** 20 points (5 pts for UHP roots, 10 pts for residues, 5 pts for final sum/evaluation)
- **Exercise 4:** 20 points (10 pts for z -substitution/expansion, 10 pts for evaluating the residue)
- **Exercise 5 (Grad):** 20 points (5 pts for poles/residues, 15 pts for sum setup and limit derivation)

Exercise 1

Consider the rational function: $f(z) = \frac{4}{(z-2)(z-6)}$

Solution: (a) **Partial Fraction Decomposition:**

$$\frac{4}{(z-2)(z-6)} = \frac{A}{z-2} + \frac{B}{z-6} \implies A(z-6) + B(z-2) = 4$$

Let $z = 2 \implies A(-4) = 4 \implies A = -1$.

Let $z = 6 \implies B(4) = 4 \implies B = 1$.

$$f(z) = -\frac{1}{z-2} + \frac{1}{z-6}$$

(b) **Series Expansions:**

i. **Region $|z| < 2$:** Both ratios $|z/2| < 1$ and $|z/6| < 1$.

$$-\frac{1}{z-2} = \frac{1}{2(1-z/2)} = \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}$$

$$\frac{1}{z-6} = -\frac{1}{6(1-z/6)} = -\sum_{n=0}^{\infty} \frac{z^n}{6^{n+1}}$$

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2^{n+1}} - \frac{1}{6^{n+1}} \right) z^n$$

ii. **Region $2 < |z| < 6$:** Here $|2/z| < 1$ and $|z/6| < 1$.

$$-\frac{1}{z-2} = -\frac{1}{z(1-2/z)} = -\sum_{n=0}^{\infty} \frac{2^n}{z^{n+1}}$$

The second term remains the same as in region (i).

$$f(z) = -\sum_{n=0}^{\infty} \frac{2^n}{z^{n+1}} - \sum_{n=0}^{\infty} \frac{z^n}{6^{n+1}}$$

iii. Region $|z| > 6$: Both ratios $|2/z| < 1$ and $|6/z| < 1$.

$$\frac{1}{z-6} = \frac{1}{z(1-6/z)} = \sum_{n=0}^{\infty} \frac{6^n}{z^{n+1}}$$

The first term remains the same as in region (ii).

$$f(z) = \sum_{n=0}^{\infty} \frac{6^n - 2^n}{z^{n+1}}$$

Exercise 2

Evaluate $\oint_{|z|=4} \frac{2z^2 - z}{(z-3i)^2(z+2)} dz$.

Solution: The integrand has two singularities inside the circle $|z| = 4$: a simple pole at $z = -2$ and a pole of order 2 at $z = 3i$.

Residue at $z = -2$:

$$\text{Res}(f, -2) = \left. \frac{2z^2 - z}{(z-3i)^2} \right|_{z=-2} = \frac{2(4) - (-2)}{(-2-3i)^2} = \frac{10}{4+12i-9} = \frac{10}{-5+12i}$$

Multiply by the conjugate to clean up:

$$\frac{10(-5-12i)}{(-5)^2+12^2} = \frac{-50-120i}{169}$$

Residue at $z = 3i$: Use the derivative limit formula for $m = 2$:

$$\text{Res}(f, 3i) = \lim_{z \rightarrow 3i} \frac{d}{dz} \left[\frac{2z^2 - z}{z+2} \right]$$

By the quotient rule:

$$\frac{d}{dz} \left[\frac{2z^2 - z}{z+2} \right] = \frac{(4z-1)(z+2) - (2z^2-z)(1)}{(z+2)^2} = \frac{2z^2+8z-2}{(z+2)^2}$$

Evaluate at $z = 3i$:

$$\text{Numerator: } 2(3i)^2 + 8(3i) - 2 = -18 + 24i - 2 = -20 + 24i$$

$$\text{Denominator: } (3i+2)^2 = -9 + 12i + 4 = -5 + 12i$$

So the residue is:

$$\frac{-20 + 24i}{-5 + 12i} = \frac{(-20 + 24i)(-5 - 12i)}{169} = \frac{100 + 240i - 120i + 288}{169} = \frac{388 + 120i}{169}$$

Total Integral: By the Residue Theorem, we sum the residues and multiply by $2\pi i$:

$$\sum \text{Res} = \frac{-50 - 120i}{169} + \frac{388 + 120i}{169} = \frac{338}{169} = 2$$

$$\oint_{|z|=4} f(z) dz = 2\pi i(2) = 4\pi i$$

Exercise 3

Evaluate the real integral: $\int_{-\infty}^{\infty} \frac{x^2}{x^4 + 1} dx$.

Solution: Let $f(z) = \frac{z^2}{z^4 + 1}$. We integrate over a large semicircular contour C_R in the upper half-plane. Solve $z^4 = -1 = e^{i\pi}$. The two roots in the UHP are:

$$z_0 = e^{i\pi/4} = \frac{\sqrt{2}}{2}(1 + i) \quad \text{and} \quad z_1 = e^{i3\pi/4} = \frac{\sqrt{2}}{2}(-1 + i)$$

Using the $P(z)/Q'(z)$ rule for the residues, $Q'(z) = 4z^3$:

$$\text{Res}(f, z_k) = \frac{z_k^2}{4z_k^3} = \frac{1}{4z_k} = \frac{1}{4}e^{-i\arg(z_k)}$$

Evaluating each:

$$\text{Res}(f, z_0) = \frac{1}{4}e^{-i\pi/4} = \frac{1}{4} \left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \right) = \frac{\sqrt{2}}{8} - i\frac{\sqrt{2}}{8}$$

$$\text{Res}(f, z_1) = \frac{1}{4}e^{-i3\pi/4} = \frac{1}{4} \left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \right) = -\frac{\sqrt{2}}{8} - i\frac{\sqrt{2}}{8}$$

Sum of the residues:

$$\sum \text{Res} = -i\frac{2\sqrt{2}}{8} = -i\frac{\sqrt{2}}{4}$$

By the Residue Theorem, the integral evaluates to:

$$\oint f(z) dz = 2\pi i \left(-i\frac{\sqrt{2}}{4} \right) = \frac{2\pi\sqrt{2}}{4} = \frac{\pi\sqrt{2}}{2}$$

The integral along the arc vanishes as $R \rightarrow \infty$ due to the degree 4 denominator vs degree 2 numerator, making this the final real answer.

Exercise 4

Use the Residue Theorem to evaluate the following definite integral:

$$\int_0^{2\pi} \sin^2(\theta) \cos^2(\theta) d\theta$$

Solution: Use the standard substitution $z = e^{i\theta} \implies d\theta = \frac{dz}{iz}$, along with $\sin \theta = \frac{z - 1/z}{2i}$ and $\cos \theta = \frac{z + 1/z}{2}$. The integral transforms into a contour integral over the unit circle $C : |z| = 1$:

$$\int_0^{2\pi} \sin^2(\theta) \cos^2(\theta) d\theta = \oint_{|z|=1} \left(\frac{z - 1/z}{2i} \right)^2 \left(\frac{z + 1/z}{2} \right)^2 \frac{dz}{iz}$$

First, combine the terms inside the squares to simplify the algebra:

$$\left(\frac{z - 1/z}{2i} \right)^2 \left(\frac{z + 1/z}{2} \right)^2 = \frac{1}{-16} [(z - z^{-1})(z + z^{-1})]^2 = -\frac{1}{16} (z^2 - z^{-2})^2$$

Now expand the binomial $(z^2 - z^{-2})^2$:

$$-\frac{1}{16} (z^4 - 2(z^2)(z^{-2}) + z^{-4}) = -\frac{1}{16} (z^4 - 2 + z^{-4})$$

Substitute this back into the contour integral:

$$\oint_{|z|=1} -\frac{1}{16} (z^4 - 2 + z^{-4}) \frac{dz}{iz} = \frac{i}{16} \oint_{|z|=1} \left(z^3 - \frac{2}{z} + \frac{1}{z^5} \right) dz$$

The only term with a non-zero residue inside $|z| = 1$ is the $1/z$ term, which has a coefficient of -2 .

$$\text{Res}(f, 0) = -2$$

Applying the Residue Theorem:

$$\int_0^{2\pi} \sin^2(\theta) \cos^2(\theta) d\theta = 2\pi i \left(\frac{i}{16} \cdot (-2) \right) = 2\pi i \left(-\frac{2i}{16} \right) = 2\pi \left(\frac{2}{16} \right) = \frac{\pi}{4}$$

Exercise 5 (Graduate Only)

Evaluate the infinite series: $\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 16}$.

Solution: Let $f(z) = \frac{\pi \cot(\pi z)}{z^2 + 16}$. We integrate over the large square contour C_N . The poles of $f(z)$ occur at $z = n$ (from $\sin(\pi z) = 0$) and at $z = \pm 4i$ (from $z^2 + 16 = 0$).

Residues at integers $z = n$:

$$\text{Res}(f, n) = \frac{\pi \cos(\pi z)}{\pi \cos(\pi z)(z^2 + 16)} \Big|_{z=n} = \frac{1}{n^2 + 16}$$

Residues at $z = \pm 4i$:

$$\operatorname{Res}(f, 4i) = \left. \frac{\pi \cot(\pi z)}{z + 4i} \right|_{z=4i} = \frac{\pi \cot(4\pi i)}{8i} = \frac{\pi(-i \coth(4\pi))}{8i} = -\frac{\pi}{8} \coth(4\pi)$$

By the symmetry of the roots and the cotangent function, $\operatorname{Res}(f, -4i) = -\frac{\pi}{8} \coth(4\pi)$ as well.

Applying the Residue Theorem:

$$\oint_{C_N} f(z) dz = 2\pi i \left[\sum_{n=-N}^N \frac{1}{n^2 + 16} - \frac{\pi}{8} \coth(4\pi) - \frac{\pi}{8} \coth(4\pi) \right]$$

Taking the limit as $N \rightarrow \infty$, the integral over C_N goes to 0:

$$0 = 2\pi i \left[\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 16} - \frac{\pi}{4} \coth(4\pi) \right]$$

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 16} = \frac{\pi}{4} \coth(4\pi)$$